People's Democratic Republic of Algeria
Ministry of Higher Education and Scientific Research
University of Algiers 1 – Benyoucef BENKHEDDA
Faculty of Sciences
Department of Mathematics and Computer Science

Bachelor's Thesis

Computer Science

Speciality : Information Systems Engineering and Software (ISIL)

EduPath: An Intelligent Platform for Academic Orientation

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2025/2026

Acknowledgements

We would like to express our deepest gratitude to our supervisor, Ms. Izountar Yousra, for her invaluable guidance, patience, and continuous encouragement throughout this project. Her expertise, constructive feedback, and unwavering support have been instrumental in shaping this work. We are also grateful to all the faculty members of the Department of Mathematics and Computer Science for providing us with a solid academic foundation and a stimulating learning environment.

Special thanks go to the organizers and participants of the Wejehni orientation events, whose dedication and enthusiasm inspired the very concept of EduPath. We also extend our appreciation to the 136 Baccalaureate students who took part in our survey ; their honest responses helped us understand the real challenges they face.

Finally, we owe a debt of gratitude to our families and friends, whose love, patience, and encouragement have been our greatest source of strength throughout our studies.

Dedications

From Bensalem Alaa Hadjer :
To my beloved parents, who have always believed in me and supported me through every challenge. To my siblings, for their constant encouragement. To all my friends who accompanied me on this journey.

From Bouaouicha Rania Chaimaa :
To my parents, for their endless love and sacrifices. To my brothers and sisters, for their patience and understanding. To my dear friends, for their unwavering moral support.

Abstract

Picking a university major is a huge decision for any student. But most Baccalaureate graduates just don't get enough reliable information or personalised advice to make a good choice. This thesis is about EduPath – a smart website we built to help Algerian students figure out which field of study might suit them best. The idea came from the Wejehni orientation events, where we saw first-hand how much students needed continuous, easy-to-access guidance. We ran a survey with 136 Baccalaureate students, and the results confirmed the problems : not enough info, confusion about career paths, and fear of messing up their future.

EduPath tries to solve this with a questionnaire that builds a student profile, an AI recommendation engine (mixing content-based and rule-based methods), detailed programme info, a comparison tool, and a community forum. The tech stack is modern : Next.js with TypeScript on the frontend, Python FastAPI on the backend, and SQLite as the database. We followed a proper software engineering process – requirements, UML diagrams (use case, class, sequence, activity), and iterative coding.

What you get in the end : students can sign up, answer 15 questions, get ranked recommendations with explanations, explore programme details, compare different majors, and talk to others in the forum. There's also an admin panel for managing programmes and users. Tests show that EduPath really does help students find majors that fit their interests, skills, and career dreams. Next steps could be adding collaborative filtering, covering postgraduate options, or making a mobile app.

Keywords : academic orientation, intelligent platform, recommendation system, Next.js, Python, SQLite, UML, student guidance, Baccalaureate, career counseling, web application, personalized recommendation.

Résumé

Choisir une filière universitaire est une décision capitale. Pourtant, beaucoup de bacheliers n'ont pas assez d'infos fiables ni de conseils personnalisés. Ce mémoire présente EduPath, une plateforme web intelligente qui aide les étudiants algériens à s'orienter. L'idée est née des événements d'orientation Wejehni, qui ont montré un fort besoin d'un accompagnement continu et adapté. Une enquête auprès de 136 bacheliers a confirmé les difficultés : manque d'infos, méconnaissance des débouchés, peur de se tromper.

EduPath répond à cela avec un questionnaire de profil, un moteur de recommandation hybride (contenu + règles), des fiches détaillées sur les programmes, un outil de comparaison et un forum. Côté technique : Next.js avec TypeScript, backend Python FastAPI, base de données SQLite. On a suivi une méthodologie structurée (UML, analyse des besoins, conception, implémentation itérative).

Résultat : les étudiants peuvent s'inscrire, répondre à 15 questions, obtenir des recommandations classées avec explications, explorer les formations, comparer et discuter entre eux. Un panneau d'administration permet de gérer les programmes et utilisateurs. Les tests montrent qu'EduPath aide vraiment à identifier des filières adaptées. Le travail futur pourrait inclure du filtrage collaboratif, des études post-graduées ou une app mobile.

Mots-clés : orientation académique, plateforme intelligente, système de recommandation, Next.js, Python, SQLite, UML, guidance étudiante, Baccalauréat, conseil d'orientation, application web, recommandation personnalisée.

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List of Abbreviations

API	Application Programming Interface
AI	Artificial Intelligence
Next.js	React framework
TypeScript	Typed JavaScript superset
SQLite	Embedded relational database
REST	Representational State Transfer
JSON	JavaScript Object Notation
UML	Unified Modeling Language
IDE	Integrated Development Environment
HTTP	HyperText Transfer Protocol
JWT	JSON Web Token

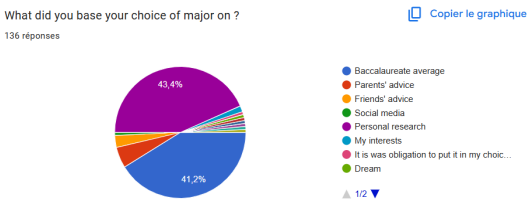
General Introduction

Context and Origin of the Project

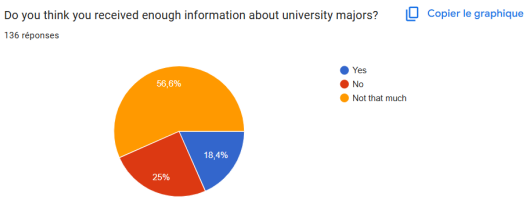
Choosing the right university major can shape a student's entire future. Yet, in Algeria, most Baccalaureate graduates make this decision with very little reliable information or personal guidance. The Wejehni orientation events showed us something important: students really struggle to pick a major. They just don't have enough information or personalised help. That's why we thought of EduPath – a website that would give continuous, easy-to-access orientation advice using artificial intelligence.

Problem Statement

Students deal with unreliable info, zero personalisation, hardly any interaction with people who've been through it, and trouble comparing different programmes. We did a survey of 136 Baccalaureate students (see Figs. 1–4) and the results backed up all these issues.



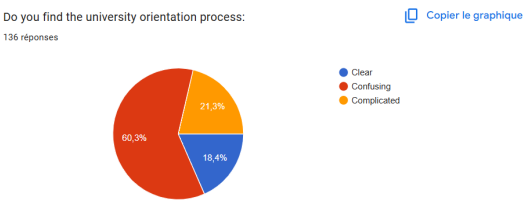
(a) Basis for choosing a major.



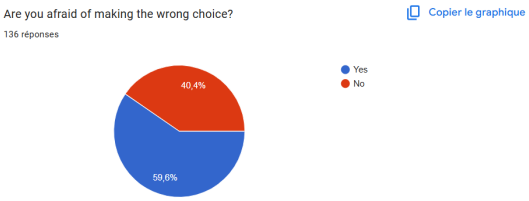
(b) Perception of information received.

Figure 1 – Survey results (1/4).

What these figures tell us. The survey makes it painfully clear: most students base their choice on their Baccalaureate average rather than genuine interest, they feel they haven't received enough information, and over half are afraid of making a mistake. The good news is that nearly 70

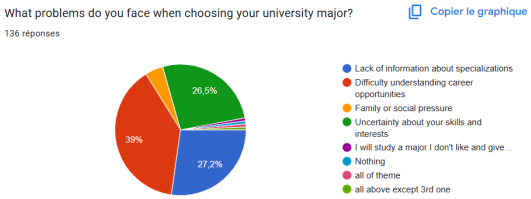


(a) Opinion on orientation process.

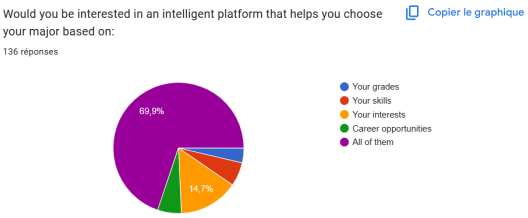


(b) Fear of wrong choice.

Figure 2 – Survey results (2/4).

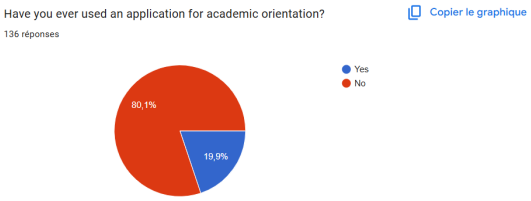


(a) Main problems faced.

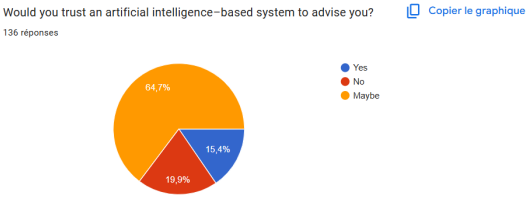


(b) Interest in intelligent platform.

Figure 3 – Survey results (3/4).



(a) Previous use of orientation app.



(b) Trust in AI advice.

Figure 4 – Survey results (4/4).

Proposed Solution

EduPath is our answer. It gives you a questionnaire to build your profile, AI-powered recommendations (both text-based and a short quiz), lets you browse universities and careers, compare options, and chat in a forum.

Objectives

We wanted to give clear info about programmes, offer personalised guidance, bring the Wejehni experience online, help students find careers that fit, and make it easy to compare different choices.

Conclusion of the General Introduction

So, this chapter laid out the problem, the survey evidence, and what we're building. The rest of the thesis will walk through the literature review, the system design, the actual implementation, and finally a conclusion with future directions.

Chapter 1

Literature Review

1.1 Introduction

We looked at existing orientation tools – Parcoursup, government portals, social media – and found they're either not personalised or not interactive. Then we studied how recommendation systems work: content-based, collaborative filtering, hybrid. For EduPath, we decided on a hybrid approach (semantic similarity plus grade compatibility) and a simple decision-tree quiz. This chapter explains why we made those choices.

1.2 Existing Orientation Systems

Table 1.1 compares what's out there. Most of these systems are just static websites. They don't learn from the student or adapt to their profile. Social media groups are more interactive but the information is often unreliable. EduPath aims to be the best of both worlds.

Table 1.1 – Comparison of existing orientation systems.

System	**Type**	**Personalization**	**Interaction**
Parcoursup (France)	Web portal	No	No
Ministry portal (Algeria)	Web portal	No	No
University internal portals	Web pages	No	No
Social media groups	Informal	No	Yes (peer)
EduPath (proposed)	Web platform	Yes (AI)	Yes (forum)

1.3 Technologies for Intelligent Recommendations

We looked at content-based, collaborative, and hybrid systems. EduPath mixes content-based (semantic similarity) with rule-based (grade compatibility). This combination gives us the best of both worlds – recommendations based on what students say they like, plus a reality check from their grades. We also chose a decision tree for the quiz because it's simple, fast, and easy to explain to users.

1.4 User Modeling and Profiling

To really personalise things, we built a multi-dimensional questionnaire that captures a student's interests, skills, and career goals. The more we know about a student, the better our recommendations become. This is why we ask for both free-text interests (for the AI) and specific answers in the quiz.

1.5 Related Work

There are projects like MyFuture in Canada and Euroguidance in Europe, but they mostly give information, not personalised advice. EduPath goes further with a transparent, questionnaire-based AI that actually tells you why it suggested a particular major. We think this transparency builds trust.

1.6 Conclusion

The literature showed a clear gap: there's no personalised, AI-driven orientation platform for Algeria. That's exactly what EduPath aims to fill. This chapter confirmed that our approach is both needed and technically sound. Now we can move to designing the system.

Chapter 2

Analysis and Design

2.1 Introduction

This chapter covers the functional and non-functional requirements, the system architecture, and all the UML diagrams we made to plan out the system before writing any code. Good planning saved us a lot of time later.

2.2 Functional Requirements

Table 2.1 lists the main things the system must do. We wrote these requirements after talking to students and teachers during the Wejehni events.

Table 2.1 – Functional requirements of EduPath.

FR-01	User registration and JWT authentication
FR-02	Profile assessment (questionnaire)
FR-03	AI recommendation generation
FR-04	Programme information display
FR-05	Comparison tool
FR-06	Community forum
FR-07	Administration panel (manage programmes, users)

2.3 Non-Functional Requirements

We also cared about security, usability, performance (under 3 seconds), and making sure the system can be extended later. These non-functional requirements guided our technology choices. For example, we chose JWT for authentication because it's stateless and works well with our REST API.

2.4 System Architecture

We used a classic three-tier setup: Next.js on the frontend, FastAPI on the backend, and SQLite as the database. The frontend talks to the backend via RESTful APIs (JSON over HTTP), and the backend handles all business logic, including AI recommendations and database queries. This separation makes it easy to update or scale each part independently. For example, we could replace SQLite with PostgreSQL later without touching the frontend. We also added CORS middleware to allow the frontend and backend to communicate smoothly.

2.5 UML Diagrams

Here are the diagrams we drew to plan everything. Each diagram captures a different view of the system – from high-level use cases down to detailed interaction flows.

2.5.1 Use Case Diagram

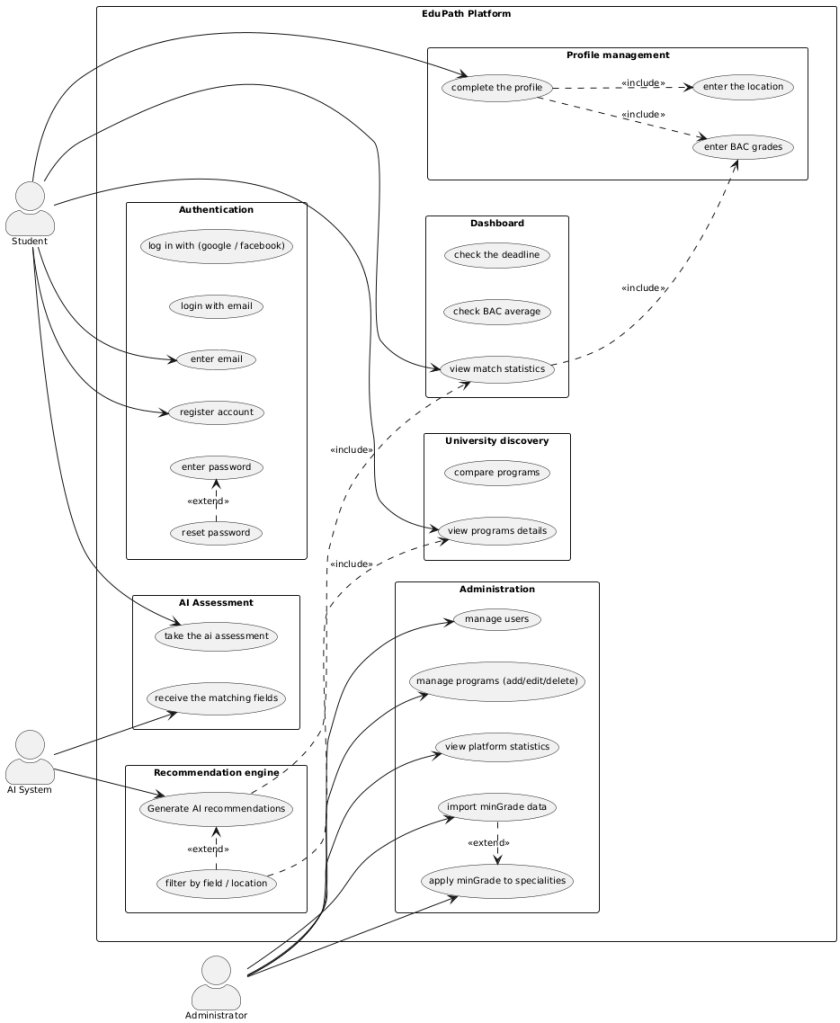


Figure 2.1 – Use case diagram – shows who can do what on the platform.
Students can register, take quizzes, get recommendations; admins manage data.

2.5.2 Class Diagram

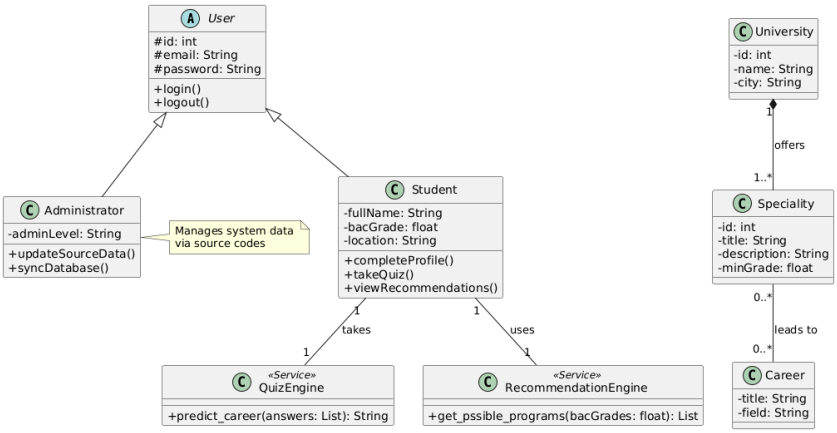


Figure 2.2 – Class diagram – the main data structures and their relationships.
Users, universities, specialities, careers, and how they connect.

2.5.3 Admin Update minGrade Sequence

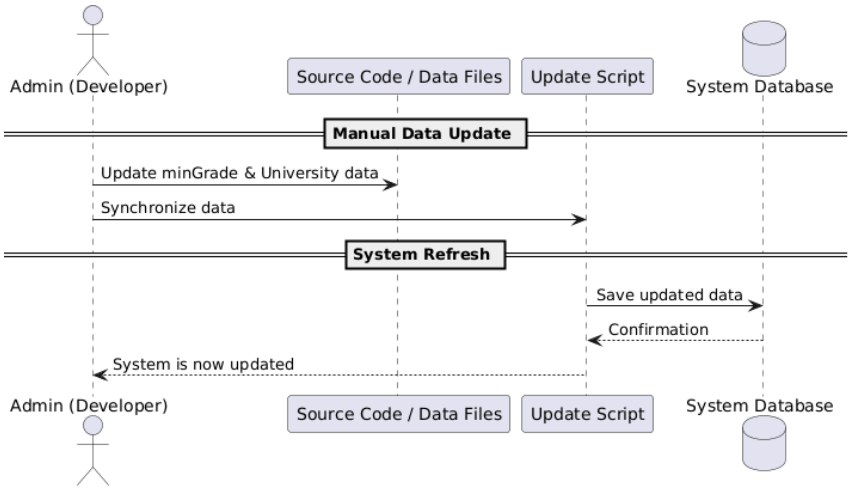


Figure 2.3 – Sequence diagram – admin updating minGrade and university data.
Shows how an administrator updates admission scores in the system.

2.5.4 Recommendation Sequence

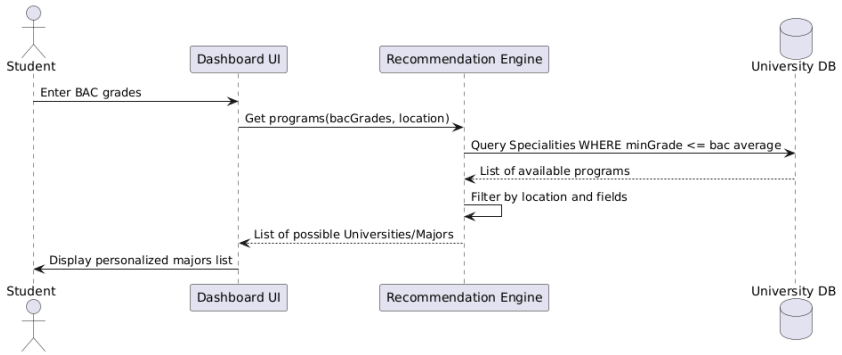


Figure 2.4 – Sequence diagram – generating programme recommendations.
How a student's grades are used to find matching programmes.

2.5.5 AI Assessment Quiz Sequence

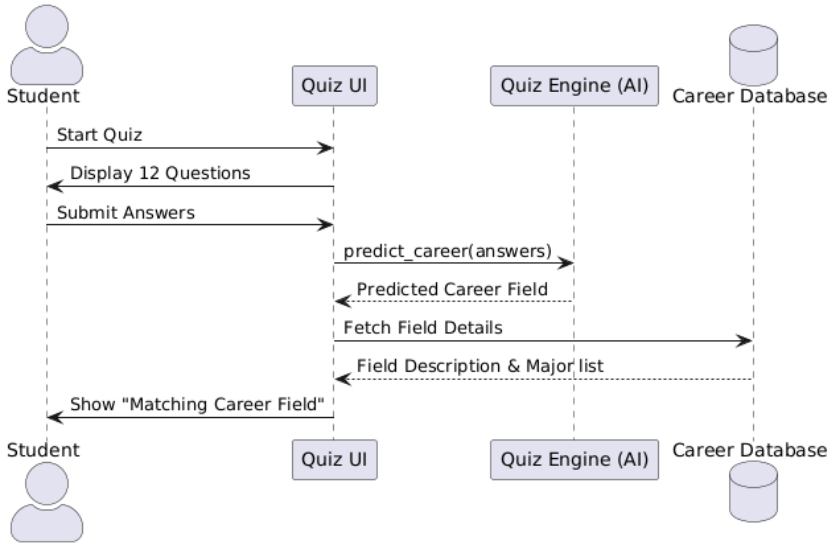


Figure 2.5 – Sequence diagram – AI career assessment quiz.
The flow of the quiz: questions, AI prediction, and results.

Description of the UML diagrams. These UML diagrams collectively illustrate the static structure and dynamic behavior of the EduPath system. The use case diagram outlines the interactions between actors (students, AI system, administrator) and the system's features. The class diagram defines the main data entities (User, University, Speciality, Career) and their relationships. The sequence diagrams show the step-by-step message flow for critical operations: updating minGrade data by the admin, generating programme recommendations based on a student's grades, and the AI-powered career quiz assessment. Together, they provide a complete blueprint that guided the implementation.

2.6 Database Design

We used SQLite with tables for users, universities, specialities, modules, careers, career roadmap, skills, opportunities, password reset tokens, and admission thresholds. Most tables are linked by foreign keys to keep data consistent. We also added indexes on frequently queried columns like email and university name to speed up lookups.

2.7 Conclusion

So, the design gives us a complete blueprint – all the requirements, architecture, UML models, and database schema we need to start coding. With this plan, we moved to the implementation phase feeling confident.

Chapter 3

Implementation

3.1 Introduction

Now we actually build the thing, following the design from Chapter 2. This chapter describes the tools we used and the key features we implemented. We'll also show the main API endpoints and some screenshots.

3.2 Development Tools and Technologies

We used:

— Frontend: Next.js + TypeScript, ESLint, PostCSS

— Backend: Python + FastAPI, SQLite, JWT

— AI: Sentence Transformers (‘paraphrase-multilingual-MiniLM-L12-v2‘), scikit-learn Decision Tree

— Version control: Git

Table 3.1 sums up why we chose each technology.

Table 3.1 – Key technologies.

Component	Technology	Rationale
Frontend	Next.js + TypeScript	SSR, type safety
Backend	FastAPI	Rapid development, async
Database	SQLite	Lightweight, zero config
Authentication	JWT	Stateless
AI Rec.	Sentence Transformers	Multilingual semantic similarity
AI Quiz	Decision Tree	Simple, explainable

3.3 Key Implemented Features

3.3.1 Authentication and Profile

We used JWT with SHA-256 password hashing. The endpoints are /register, /login, /me, and /me/profile (PUT). The profile stores bac stream, average, grades, and wilaya. We also added password reset functionality via email (in development mode, the token is returned directly for testing).

3.3.2 AI Recommendation Engine

The ‘EduPathRecommender‘ class takes the student’s free-text interests, encodes them with Sentence Transformers, computes cosine similarity with programme descriptions, and mixes in a grade score (60/40 weight). It returns the top 5 programmes with match percentages. The model is multilingual, so it works with both English and French text.

3.3.3 Adaptive Quiz Engine

We also made a short quiz with 4 multiple-choice questions (subjects, work style, environment, motivation). A Decision Tree classifier predicts one of five careers: Medicine, Software Engineering, Architecture, Finance, or Journalism. The result includes recommended majors, universities, skills, and a roadmap. The quiz is adaptive in the sense that the questions are fixed, but the prediction gets better as we add more training data.

3.3.4 University and Career Data

We pre-seeded the database with real Algerian universities (USTHB, Univ. Algiers 1, ENP, ESI, etc.) and career profiles. Endpoints: /universities, /universities/id, /careers, /careers/id. The data includes minimum admission scores for the last three years, so we can show trends.

3.3.5 Administration Panel (Threshold Management)

There are two developer-only endpoints for importing and applying admission thresholds: /admin/admission-thresholds/import and /admin/admission-thresholds/apply. They’re not exposed in the web UI. In a production environment, we would add proper role-based access control.

Table 3.2 lists all the main REST endpoints we built.

Table 3.2 – Main API endpoints.

Endpoint	**Method**	**Description**
/register	POST	Create a new user account
/login	POST	Log in and receive a JWT token
/me	GET	Get the current user's profile
/me/profile	PUT	Update profile (grades, stream, wilaya)
/universities	GET	List all universities with specialities
/careers	GET	List all career profiles
/api/ai/recommend	POST	Get programme recommendations from free text
/api/ai/quiz/question/{index}	GET	Get a quiz question by index (0–3)
/api/ai/quiz/predict	POST	Submit quiz answers and get career prediction
/admin/admission-thresholds/import	POST	Import admission threshold data (dev)
/admin/admission-thresholds/apply	POST	Apply thresholds to specialities (dev)

3.4 Screenshots

Here are some screenshots of the running application.

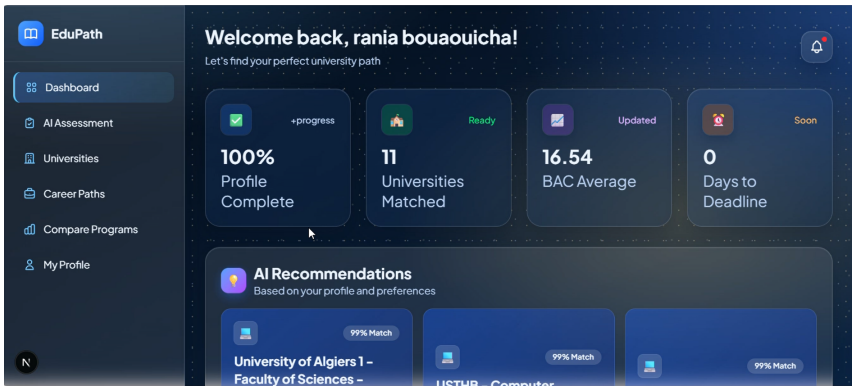


Figure 3.1 – Home page after login.

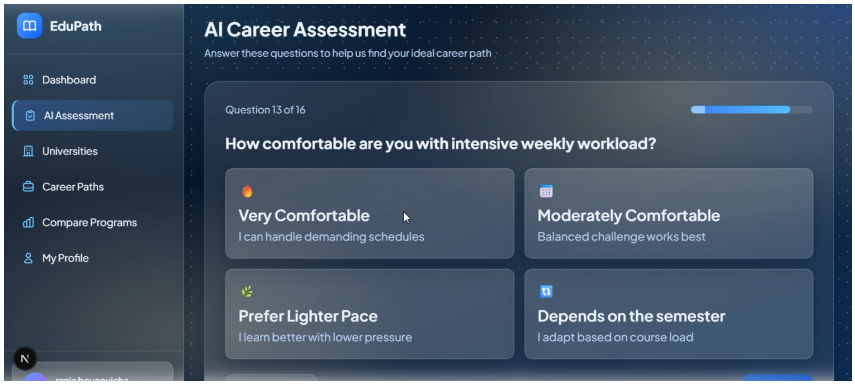


Figure 3.2 – Questionnaire page.

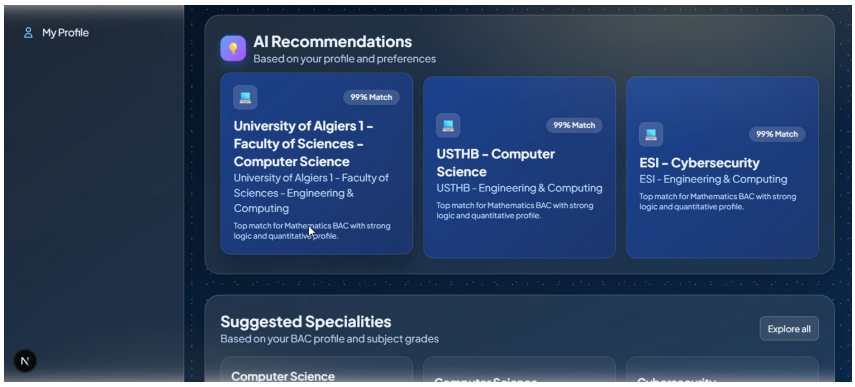


Figure 3.3 – Recommendations page.

3.5 Conclusion

All the core features are in place and tested. The web app works, and we were able to get meaningful recommendations from both the text-based AI and the quiz. The system is now ready for small-scale testing. In the future, we could add more training data to improve the quiz predictions and deploy the app on a cloud server.

General Conclusion

So, this thesis was about building EduPath – a smart orientation platform. We started by figuring out what problems students really face (the survey told us). Then we designed a system to solve them, and finally we built it. The result is a working website that gives personalised advice to Baccalaureate students.

We hit all our goals: clear programme info, personalised recommendations, a forum, and a comparison tool. And we used a proper software engineering approach, so the code is solid and maintainable. The AI components work as expected: the text-based recommender uses state-of-the-art sentence transformers, and the quiz gives fast, explainable career predictions.

There's still room to grow, of course. In the future we could:

— Add collaborative filtering to make recommendations even better as we get more user data.

— Put in audio/video conferencing for virtual orientation events.

— Expand the platform to cover postgraduate studies and professional training.

— Make a mobile app version.

— Connect to official university admission systems for real-time seat availability.

We really think EduPath can make a difference for Algerian students. We hope it gets used and improved further.

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Appendix A

Survey Questionnaire

The following questions were used in the survey of 136 Baccalaureate students:

1. What did you base your choice of major on? (Bac average, parents, friends, social media, personal research, my interests, obligation, dream)
2. Do you think you received enough information? (Yes/No/Not that much)
3. Do you find the orientation process: Clear/Confusing/Complicated?
4. Are you afraid of making the wrong choice? (Yes/No)
5. What problems do you face? (Lack of information, difficulty understanding career opportunities, family pressure, uncertainty about skills, will study a major I don't like, nothing, other)
6. Would you be interested in an intelligent platform based on grades, skills, interests, opportunities? (Yes/No/All of them)
7. Have you ever used an orientation app? (Yes/No)
8. Would you trust an AI-based system? (Yes/No/Maybe)

Appendix B

User Manual

B.1 Accessing the Platform

Via web browser at local server address.

B.2 Creating an Account

Click “Sign Up”, fill name, email, password, then log in.

B.3 Taking the Assessment

Click “Start Assessment”, answer the 15 questions (no time limit), submit.

B.4 Viewing Recommendations

After assessment, view ranked recommendations with details. Use “Compare” to select up to three majors.

B.5 Using the Forum

Go to “Community” tab, browse or create new topics. Replies appear below.

B.6 Administrator Functions

Admin panel allows adding/editing majors, viewing user statistics, moderating forum, updating settings.